

AI CITY CHALLENGE 2026 TRACK 3

Anomalous Events in Transportation

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Objective

- **Objective:** Build a unified video understanding model or system to detect, reason about, and explain anomalous events in transportation surveillance.
- **Key Requirements:** Perform diverse reasoning tasks ranging from simple event verification to open-ended causal analysis.
- **Core Mechanism:** Requires explicit chain-of-thought reasoning grounded in temporal and spatial evidence.

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Task Taxonomy

10 Tasks Across 3 Main Groups

- **Basic Group (QA & Verification):**

- Event Verification (with and without explanation)
- Multiple-Choice QA (with and without explanation)
- Open-Ended QA

- **Scene Group (Context):**

- Scene Description (Static)
- Video Summarization (Dynamic)

- **Temporal Group (Analysis):**

- Temporal Localization (When the anomaly occurred)
- Causal Linkage (What caused the anomaly)
- Event Description (What happened in the interval)

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Training Data

- **Volume:** 44,040 annotations across 3,670 CCTV transportation videos (~26.1 hours total).
- **Data Sources:** Aggregated from 8 public open-source datasets (e.g., VAD-R1, TAD, Accident-Bench, UCF Crime).
- **Auto-Labeling Pipeline:**
 - **Gemini 3.1 Pro:** Used for video captioning and structured event descriptions.
 - **Gemma-4:** Used for multi-task Q&A with chain-of-thought reasoning.
- **Human Context:** 910 videos utilize existing NVIDIA human annotations (bounding boxes, timestamps) as supplementary context.

Training Data

```
train/
├─ bcq.json                - binary Yes/No questions (7,340 items)
├─ bcq_openended.json     - binary Yes/No + explanation (7,340 items)
├─ mcq.json               - multiple choice (3,670 items)
├─ mcq_openended.json     - multiple choice + explanation (3,670 items)
├─ open_qa.json           - open-ended QA (3,670 items)
├─ scene_description.json  - scene description (3,670 items)
├─ video_summarization.json - video summary (3,670 items)
├─ temporal_localization.json - when did the anomaly occur (3,670 items)
├─ temporal_description.json - what happened in the interval (3,670 items)
└─ causal_linkage.json     - what caused the anomaly (3,670 items)
```

Training Data

```
{  
  "format": "tao-v1-reason-v1.0",  
  "metadata": {  
    "type": "annotation",  
    "task": "<task type, e.g. \"bcq\">",  
    "date": "2026-05-02",  
    "description": "...",  
    "license": "CC-BY-4.0"  
  },  
  "media_root": null,  
  "items": [ /* per-item records */ ]  
}
```

Training Data

```
{  
  "video_id": "TAD/01_Accident_001.mp4",  
  "question": "Does a rear-end collision occur in the video?\nAnswer with Yes or No.",  
  "answer": "Yes",  
  "reasoning": "The video shows a nighttime scene at a four-way intersection..."  
}
```

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The Test Split

- **Volume:** 960 human-curated annotations covering 80 short clips.
- **Source Material:** Clips are trimmed from 17 public YouTube videos, ensuring a distinct distribution from the training set.
- **Structure:** Retains the exact same 10 task types as the training data (spanning basic QA, scene understanding, and temporal reasoning).
- **Format:** Provided as a single combined file (`test.json`), but answers are redacted for the public leaderboard.
- **Data Access:** The `download_test_videos.py` script automatically uses `yt-dlp` and `ffmpeg` to fetch and trim the source clips.

The Test Split

```
{
  "format": "tao-v1-reason-v1.0",
  "metadata": {
    "type": "annotation",
    "task": "tar_test",
    "description": "AI City Challenge 2026 Track 3 – TAR test set (in-domain, 80 videos, 10 task types). Answers redacted; submit predictions to the evaluation server.",
    "license": "CC-BY-4.0",
    "answers_redacted": true
  },
  "media_root": null,
  "items": [
    {
      "item_index": "bfaa0b67a0385860",
      "task_type": "bcq",
      "video_id": "tar_test/v=-3nw0fm1Pdk_0-00_0-16.mp4",
      "question": "Does the white van flip over after the collision?\n\nAnswer with only Yes or No.",
      "answer": ""
    },
    {
      "item_index": "41dcf77f800ebcae",
      "task_type": "bcq",
      "video_id": "tar_test/v=-3nw0fm1Pdk_0-00_0-16.mp4",
      "question": "Is there a collision between two white vans?\n\nAnswer with only Yes or No.",
      "answer": ""
    },
    {
      "item_index": "b944ca5ad1567362",
      "task_type": "bcq_openended",
      "video_id": "tar_test/v=-3nw0fm1Pdk_0-00_0-16.mp4",
      "question": "Does the white van flip over after the collision?\n\nAnswer with Yes or No, followed by a brief explanation.",
      "answer": ""
    },
    {
      "item_index": "51350fb685ecb0d0",
```

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Submission Format & Local Validation

- **Submission Format:** A single CSV file with exactly two columns:
 - `item_index`: The 16-hex sample ID from the test JSON.
 - `prediction`: The model's raw output text (multi-line predictions allowed).
- **Row Requirement:** Exactly 960 rows matching the test set IDs.
- **Local Validation:**
 - The provided `evaluate.py` script runs in validation-only mode.
 - It verifies CSV structure, completeness, and task-specific parseability (e.g., verifying Yes/No, single letter MCQ, or valid JSON for temporal boundaries).

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Evaluation & Generalization

Rigorous Testing Methodology

- **In-Domain Test Set:** Human-verified traffic anomaly videos, evaluated on the exact same 10 task types as the training set.
- **Out-of-Domain Set 1 (Fisheye):** Fisheye intersection traffic monitoring footage (tests generalization to a different task formulation).
- **Out-of-Domain Set 2 (Egocentric):** Dashcam video targeting pedestrian crossing intent prediction (tests generalization across visual domains and tasks).

Evaluation Metrics

Task-Specific Scoring

Task Type	Metric
Event Verification (bcq)	Yes/No accuracy (regex extraction)
Multiple-Choice QA (mcq)	Letter accuracy (regex extraction)
Temporal Localization	Mean IoU over {"start", "end"} JSON bounds
Open-Ended / Explanatory Tasks	BERTScore F1 (roberta-large, rescaled)
Overall Score	Unweighted mean of all per-task metrics

**Open-ended tasks include open QA, causal linkage, scene description, video summary, and QA with explanations.*

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Proposed Architecture: DINOv3-based Video-LLM

A Unified Perception-Reasoning Pipeline

1. Spatial Perception (The Eyes)

- **Backbone:** DINOv3 (ViT-L/G).
- **Strength:** Dense, self-supervised spatial features for zero-shot anomaly detection.
- **Extract:** High-res patch embeddings per frame.

2. Temporal Bridge

- **Mechanism:** Perceiver Resampler or Temporal Transformer.
- **Context:** Aggregates frame-level DINO features into spatio-temporal video tokens.

3. Reasoning Engine (The Brain)

- **LLM Integration:** LLaVA-style projector maps tokens to LLM space (e.g., Llama 3 or Qwen2).
- **CoT Output:** Fine-tuned to generate *Reasoning* before *Answer*.

Task-Specific Heads

- **Autoregressive:** QA, MCQ, Descriptions, and Causal analysis.
- **Regression Head:** MLP-based 1D head for precise **Temporal Localization**.